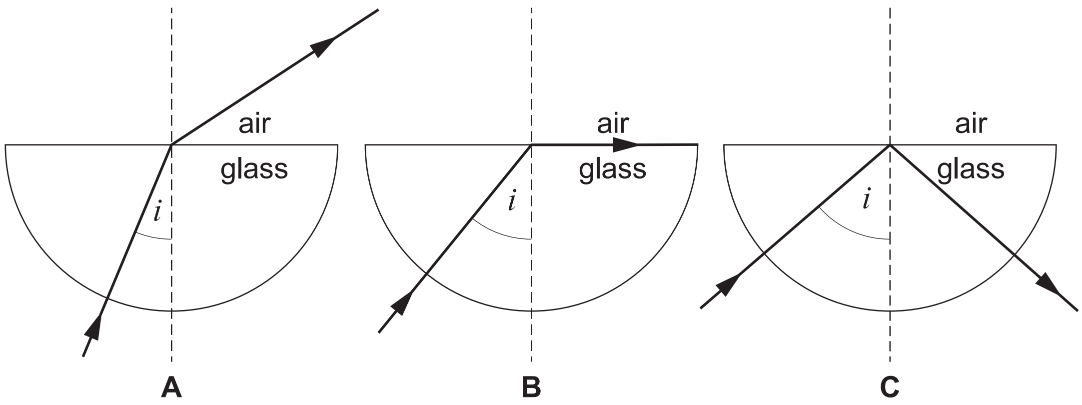
Examiner only

3. A class investigates the total internal reflection (TIR) of light in a glass block.

They change the angle of incidence, i, of a ray of light and draw in the path of the ray out of the block.

Their results are shown in the three diagrams, **A**, **B** and **C** below.



(a) (i) State in which diagram, **A**, **B** or **C**, the angle of incidence is equal to the critical angle. [1]

Diagram ....

(ii) State which diagram, **A**, **B** or **C**, shows total internal reflection. [1]

Diagram ....

(b) Tick (✓) the box next to the condition required for total internal reflection to occur. [1]

Light must be travelling from a less dense to a more dense material.	☐
Light must be travelling from one material to a different material of equal density.	☐
Light must be travelling from a more dense to a less dense material.	☐

(c) State **one** use of total internal reflection. [1]

....

4

© WJEC CBAC Ltd. (3420U10-1)